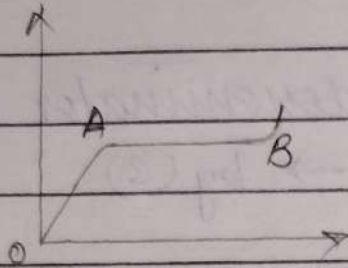


Electronics Assignment-2.

Q.1.



JFET as a VVR -

In O-A region, JFET behaves like a VVR

because in OA region, as voltage V_{DS} increases, the current I_D also increases, therefore resistance of JFET decreases, thus in OA region the resistance of JFET varies with the original applied voltage, hence can be used as a VVR.

JFET as a constant current source -

In A-B region, JFET behaves like a constant current source because as voltage V_{DS} increases the output current I_D remains constant.

Q.2 common base (CB) configuration,
current gain is $\alpha = I_C / I_E$ — (1)
common-emitter (CE) configuration,
current gain is $\beta = I_C / I_B$ — (2)

We know, $I_E = I_C + I_B$

$$1 = (I_C + I_B) / I_E$$

$$= \frac{I_C}{I_E} + \frac{I_B}{I_E} = \alpha + \frac{I_B}{I_E} \rightarrow \text{by (1)}$$

$$1 - \alpha = I_B / I_E \quad \text{--- (3)}$$

Now, $\beta = I_C / I_B \neq$

Dividing numerator & denominator by I_E

$$\Rightarrow \beta = \frac{I_C / I_E}{I_B / I_E} = \frac{\alpha}{1 - \alpha} \rightarrow \text{by (3)}.$$

$$\Rightarrow \beta = \frac{\alpha}{1 - \alpha}, \quad \alpha = \frac{\beta}{\beta + 1}$$

Given, $I_E = 10 \text{ mA}$, $\alpha = 0.987$.

To find, I_C ? I_B ?

We know, $\alpha = I_C / I_E$

$$\Rightarrow I_C = \alpha \cdot I_E = 0.987 \times 10 \\ = 9.87 \text{ mA}$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.987}{1 - 0.987} = \frac{0.987}{0.013} = 75.923$$

$$I_B = I_E - I_C = 10 - 9.87 = 0.13 \text{ mA} \\ = 130 \mu\text{A}$$

Q 3

$$\alpha_1 = 0.980, \quad \alpha_2 = 0.989$$

$$\beta_1 = \frac{\alpha_1}{1 - \alpha_1} = \frac{0.980}{1 - 0.980} = \frac{0.98}{0.02} = 49$$

$$\beta_2 = \frac{\alpha_2}{1 - \alpha_2} = \frac{0.989}{1 - 0.989} = \frac{0.989}{0.011} = 89.9091$$

$$\% \text{ change} = \frac{\beta_2 - \beta_1}{\beta_1} \times 100$$

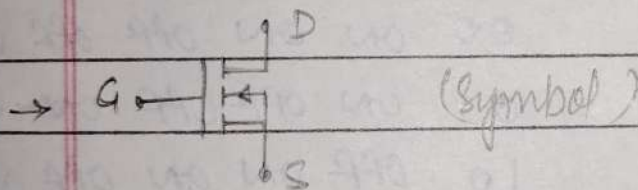
$$= \frac{89.9091 - 49}{49} \times 100 = \frac{40.9091}{49} \times 100$$

$$= 83.49\%$$

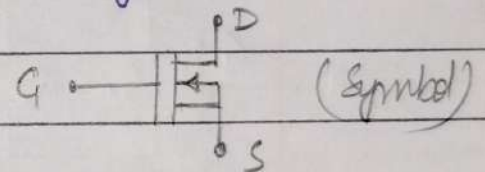
Q.4. E-MOSFET

D-MOSFET.

- There is no channel in EMOS, channel needs to be induced.



- There is a channel in DMOS without inducing.

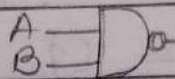


Q.5. BJT

FET.

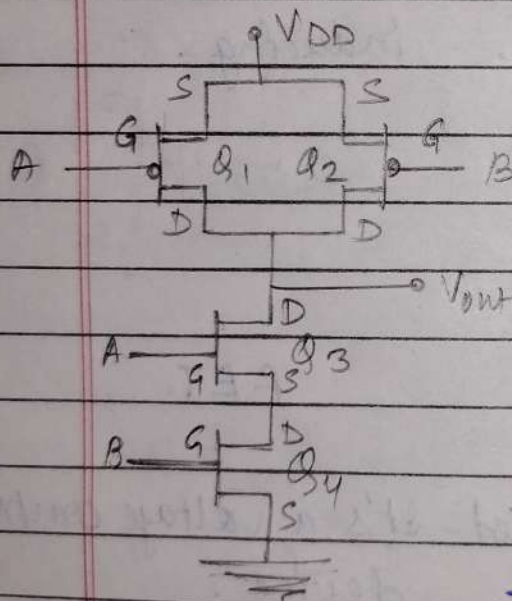
- It's a current controlled device.
- uses both, electrons & holes.
- 3 terminals of BJT are Emitter, Base, Collector
- Types of BJT are - pnp & npn BJTs.
- It's a voltage controlled device.
- uses only one type of carriers (electrons/holes)
- 3 terminals are Source, Drain, Gate.
- Types of FET are - JFET, MESFET, MOSFET.

Q 5. CMOS as NAND Gate -

NAND Gate -  $y = \overline{A \cdot B}$

Truth Table

A	B	$y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0



PMOS

AB	Q ₁	Q ₂	Q ₃	Q ₄	V _{out}
00	ON	ON	OFF	OFF	V _{DD} (1)
01	ON	OFF	OFF	ON	V _{DD} (1)
10	OFF	ON	ON	OFF	V _{DD} (1)
11	OFF	OFF	ON	ON	GND (0)

When $Q_1 = Q_2 = Q_3 = Q_4 = 0$,
PMOS get short, NMOS get off.

⇒ $V_{out} = V_{DD}$

When $Q_1 = Q_3 = 0$, $Q_2 = Q_4 = 1$,
 Q_1 & Q_4 are ON, Q_2 & Q_3 are OFF.

⇒ $V_{out} = V_{DD}$

When $Q_1 = Q_3 = 1$, $Q_2 = Q_4 = 0$,
 Q_1 & Q_4 are off, Q_2 & Q_3 are ON

⇒ $V_{out} = V_{DD}$

When $Q_1 = Q_2 = Q_3 = Q_4 = 1$,
NMOS get short, PMOS get off

⇒ $V_{out} = \text{Ground}$